Answer key: Lesson 9, Draw and Measure

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. In the box, draw a line 7 cm long. Start at the 0 mark.

Answer: A line from the 0 mark to the 7 mark: it starts at 0, ends at the 7 mark and is 7 cm long

Put a dot for the start. Line up the 0 mark with the dot. Make a small mark at the 7. Join the dot and the mark with a straight line. The line ends at the 7 mark, so it is 7 cm long.

Watch for: Starting at the edge of the ruler instead of the 0 mark (the line comes out a little too long).; Drawing to the 8 mark because the 0 was counted as one.

S2. Draw a line 4 cm long. Under it, draw a line 9 cm long.

Answer: 9 cm; it is 5 cm longer

The 9 cm line is the longer one. To find how much longer, count on from 4 to 9: that is 5 jumps. So it is 5 cm longer (9 take away 4 is 5).

Watch for: Writing 13 by adding the two lengths.; Starting the 9 cm line at a mark other than 0.

S3. Measure the length of your pencil. Write your guess first. Then measure to the nearest cm.

Answer: OPEN ANSWER, accept any sensible measure (guess and measure within about 3 cm of each other); a new pencil is about 17 to 19 cm. Check: the 0 mark at one end, read the mark at the other end

Put one end of the pencil on the 0 mark and keep the ruler straight. Read the mark at the other end. A new pencil is usually between 17 and 19 cm. If the tip is between two marks, write the closer one.

Watch for: Starting at the edge of the ruler instead of the 0 mark.; Writing the guess in the measure space, or leaving off the unit.

S4. Ali's pencil and Mei's pencil both start at the 0 mark of a ruler. Ali's tip is at the 13 mark. Mei's tip is at the 9 mark.

Answer: 13 cm; 9 cm; Ali's is 4 cm longer

Both start at the 0 mark, so each tip mark is the length: Ali 13 cm, Mei 9 cm. From 9 to 13 is 4 jumps (13 take away 9 is 4), so Ali's pencil is 4 cm longer.

Watch for: Adding 13 + 9 = 22.; Writing 9 as the difference (that is the length of Mei's pencil).

S5. Draw a line 5 cm long. Under it, draw a line 3 cm longer than the first line.

Answer: The second line is 8 cm long (5 + 3)

The first line is 5 cm. 3 cm longer is 5 + 3 = 8 cm. Draw it from the 0 mark to the 8 mark, under the first line.

Watch for: Drawing a line 3 cm long.; Starting the second line at the edge of the ruler, so it is a little too long.

S6. Mei's line goes from the 0 mark to the 7 mark. Mei says it is 8 cm long.

Answer: No. Her line is 7 cm. She should draw on to the 8 mark

Mei started at 0 and stopped at the 7 mark, so her line is 7 cm long. A line 8 cm long must end at the 8 mark. She should draw on to the 8 mark.

Watch for: Saying yes because Mei said 8.; Counting the 0 as one, so the 7 mark looked like 8.

S7. The tip of Ben's spoon is between the 14 mark and the 15 mark. It is closer to the 15 mark. The spoon starts at the 0 mark.

Answer: 15 cm (the tip is closer to the 15 mark)

The tip is between 14 and 15 and it is closer to 15. So, to the nearest centimetre, the spoon is 15 cm long.

Watch for: Writing 14 because the tip has not reached 15.; Writing 14 and a half.

Think harder (page 3)

T1. Use your ruler to draw a path with 3 straight pieces: 5 cm, 3 cm and 4 cm.

Answer: 12 cm (5 + 3 + 4)

Draw 5 cm, turn, draw 3 cm, turn, draw 4 cm. The path is as long as all three pieces together: 5 + 3 = 8, and 8 + 4 = 12 cm.

Watch for: Measuring in a straight line from the start to the end of the path.; Drawing one long line instead of turning at the corners.

T2. Line A starts at the 2 mark and ends at the 9 mark. Line B starts at the 3 mark and is 6 cm long.

Answer: 7 cm; the 9 mark

(a) From the 2 mark to the 9 mark is 7 jumps: 9 take away 2 is 7 cm, not 9 cm. (b) Start at the 3 mark and jump on 6: 3 + 6 = 9. The line ends at the 9 mark.

Watch for: Writing 9 for line A because the end is at the 9 mark.; Taking 6 away from 3, or writing 6 for line B.

T3. Draw a line 13 cm long in the box. Put a dot on it 5 cm from the start.

Answer: 8 cm (13 take away 5)

The whole line is 13 cm and the part before the dot is 5 cm. The rest is 13 take away 5, which is 8 cm. Check by measuring from the dot to the end of the line.

Watch for: Writing 5, the part that is already known.; Adding 13 + 5 = 18.

T4. Both pencils start at 0. Ali's tip is between 16 and 17, closer to 16. Mei's tip is between 18 and 19, closer to 19.

Answer: 16 cm; 19 cm; 3 cm longer

Ali's tip is closer to 16, so 16 cm. Mei's tip is closer to 19, so 19 cm. From 16 to 19 is 3 jumps, so Mei's pencil is 3 cm longer.

Watch for: Choosing the mark the tip has already passed (17 and 18).; Writing 2 by using 17 and 19.

Go further (page 4)

G1. A ribbon is 30 cm long. Mei cuts all of it into 3 pieces: A, B and C. Piece A is 8 cm long. Piece B is 3 cm longer than piece A. How long is piece C?

Answer: Piece C is 11 cm (B is 11 cm, A and B together are 19 cm, and 30 take away 19 is 11)

B is 8 + 3 = 11 cm. A and B together are 8 + 11 = 19 cm. The ribbon is 30 cm, so C is 30 take away 19, which is 11 cm. Check: 8 + 11 + 11 = 30.

Watch for: Taking 30 - 8 = 22 and stopping.; Writing 3 for piece B instead of 3 more than A.

G2. A snail starts at the 4 mark of a long ruler. It crawls to the 13 mark. Then it crawls back 5 cm towards the 0 mark.

Answer: At the 8 mark; it crawled 14 cm in all (9 cm forward and 5 cm back)

Forward from the 4 mark to the 13 mark is 9 cm. Back 5 cm: 13 take away 5 puts the snail at the 8 mark. In all it crawled 9 + 5 = 14 cm, although it is only 4 cm from where it began.

Watch for: Writing 9 for the total and forgetting the way back.; Writing 4 as the mark (the distance from the start, not the mark).

G3. Draw a path with 3 straight pieces. The pieces are all different lengths, in whole cm. The whole path is exactly 20 cm long. Many answers are right.

Answer: OPEN ANSWER, accept any 3 different whole numbers that add up to 20, for example 5, 7 and 8 (5 + 7 + 8 = 20)

One answer is 5, 7 and 8: 5 + 7 is 12 and 12 + 8 is 20. Other answers are 3, 8 and 9, or 4, 7 and 9. If the total is not 20, change one piece until it is.

Watch for: Two pieces the same length (for example 5, 5 and 10).; A total of 19 or 21 because a piece was drawn a little too long or too short.

Investigation (page 5): Does the start mark matter?

The length stays 6 cm every time. From the 2 mark the line ends at 8; from 3 it ends at 9; from 5 it ends at 11. The end mark is always the start mark plus 6, so the end mark take away the start mark is always 6. Reading only the end mark gives the length only when the line starts at 0.

Look for: A child who tries more than one start mark, writes both marks, and says that the end mark take away the start mark (or the number of jumps) is the length. A 15 cm ruler allows start marks up to 9.

Exit ticket (page 5)

E1. Draw a line 6 cm long on your spare paper. Start it at the 0 mark of your ruler. Then fill in the marks.

Answer: A line from the 0 mark to the 6 mark

Put the 0 mark of the ruler on the dot and draw to the 6 mark. The line starts at 0 and ends at 6.

Watch for: Starting at the edge of the ruler.; Ending at the 7 mark after counting the 0 as one.

E2. Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.

Answer: 9 cm longer

The rubber is 6 cm and the pencil is 15 cm. 15 take away 6 is 9, so the pencil is 9 cm longer.

Watch for: Adding: 21.; Writing 15 (the pencil, not the difference).

E3. A line starts at the 3 mark and ends at the 10 mark. How long is the line?

Answer: 7 cm. 10 take away 3 is 7 (7 jumps from 3 to 10)

Count the jumps from 3 to 10: 7 jumps. Or take 10 take away 3, which is 7. The line is 7 cm long, not 10 cm.

Watch for: Writing 10 (the end mark).; Writing 13 by adding.

Sneak-peek puzzles (not in the workbook)

1. A line from the dot that is 3 cm long: the 0 mark on the dot and the 3 mark at the other end.
2. No. It is 8 cm long. The pencil starts at the 1 mark, not the 0 mark, so the first centimetre of the ruler is not part of the pencil (9 take away 1 is 8).